Thurrja

Sistemet kontinuale



$$y(t) = x(t) * h(t)$$

$$x(t) * \delta(t) = x(t)$$

$$x(t) * \delta(t-t_0) = x(t-t_0)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

1° "$t \to \tau$"

2° Pasqyrim "$-\tau$"

3° Zhvendosja për "t" mayitas $|t-\tau|$

1. Vetia asociative

$$x(t) * \{h_1(t) * h_2(t)\} = \{x(t) * h_1(t)\} * h_2(t)$$

2. Vetia komutative

$$x(t) * h(t) = h(t) * x(t)$$

3. Vetia distributive

$$x(t) * \{h_1(t) * h_2(t)\} = x_1(t) * h_1(t) + x(t) * h_2(t)$$

4. Vetia e zhvendosjes $y(t-t_0) = x(t-t_0) * h(t) = x(t) * h(t-t_0)$

Detyra

2.4 Janë dhënë hyrja $x(t)$ dhe përgjigjja impulsive $h(t)$ si në vijim:

$$x(t) = u(t)$$

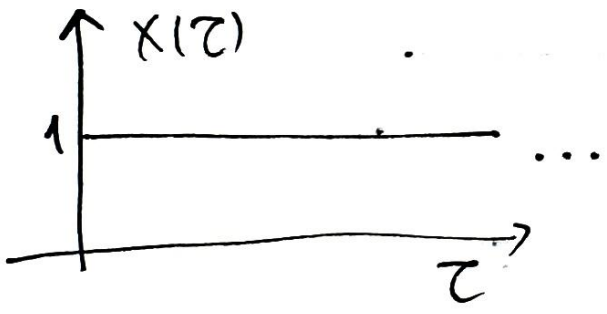
$$h(t) = e^{-\alpha t} u(t) \qquad \alpha > 0$$

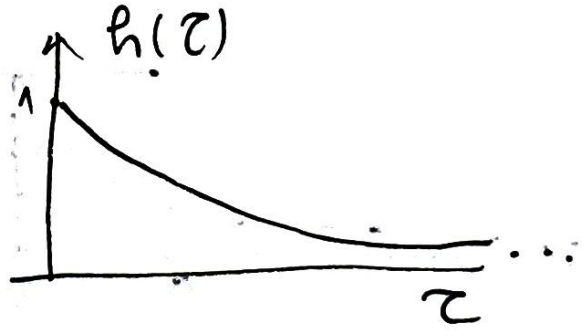
Gjeni:

a) $y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$

b) $y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$

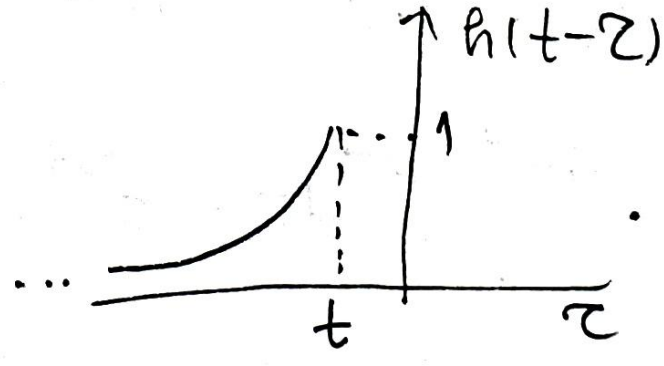
Zgjidhje





Pasqyrojm dhe zhvendosim $h(\tau)$:

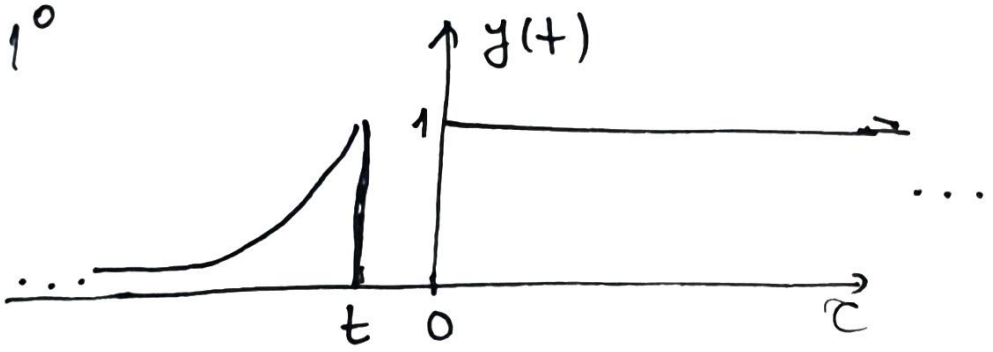




$$x(\tau) = u(\tau)$$

$$h(t-\tau) = e^{-\alpha(t-\tau)} u(t-\tau)$$

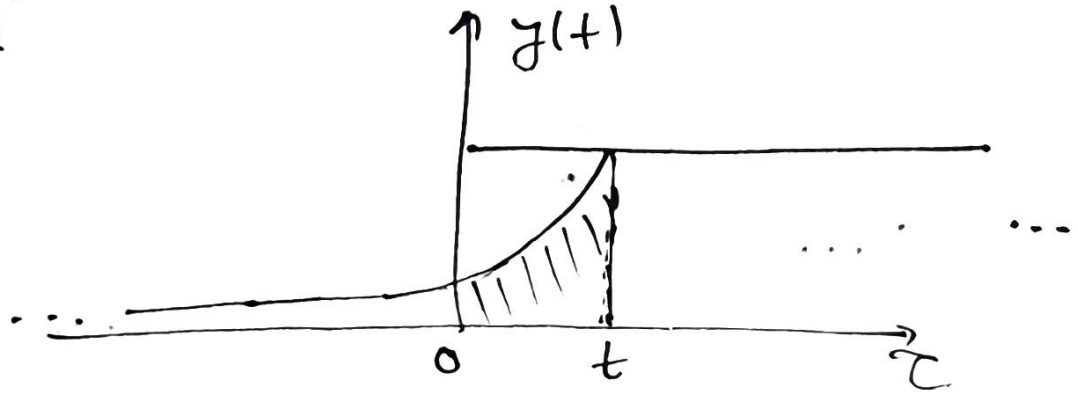
1°



$t<0$

$$y(t) = \int_{-\infty}^{t} x(\tau) h(t-\tau) d\tau = 0 \rightarrow \text{Nuk ka thurrje}$$

2°



$t>0 \quad 0<t$

$$y(t) = \int_{0}^{t} 1 \cdot e^{-(t-\tau)} d\tau = \int_{0}^{t} e^{-t} \cdot e^{\tau} d\tau$$

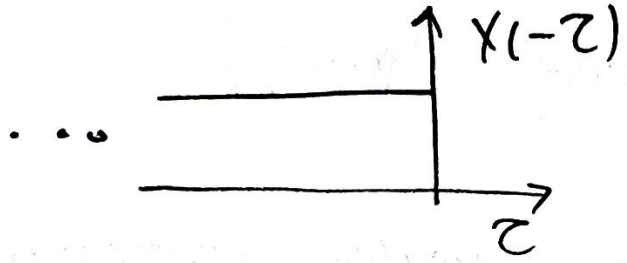
$$= e^{-\alpha t} \int_{0}^{t} e^{\alpha\tau} d\tau = e^{-\alpha t} \frac{1}{\alpha} e^{\alpha\tau} \Big|_{0}^{t} = e^{-\alpha t} \frac{1}{\alpha} \left(e^{\alpha t} - e^{0}\right) =$$

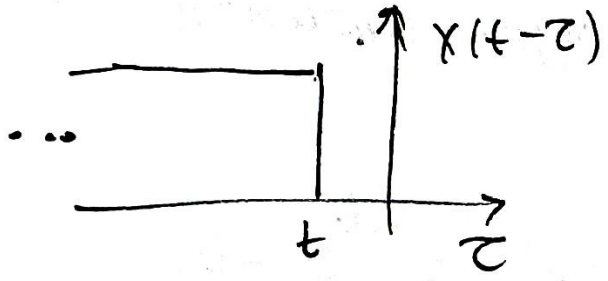
$$= e^{-\alpha t} \frac{1}{\alpha} \left(e^{\alpha t} - 1\right) = \frac{1}{\alpha} \left(1 - e^{-\alpha t}\right).$$

b) $y(t) = h(t) * x(t)$

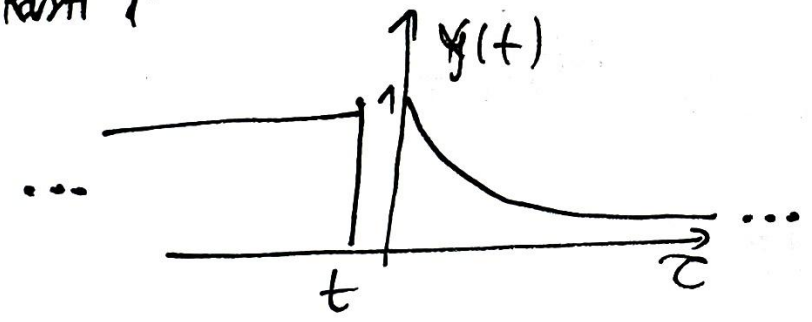
$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

Pasqyrojmë dhe zhvendosim $x(\tau)$





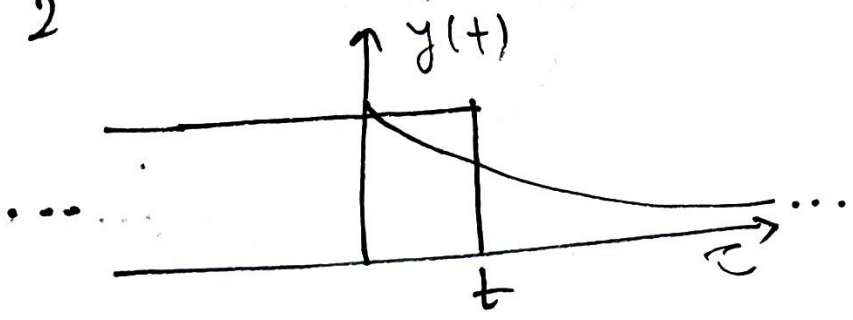
Rasti 1°



1° $t < 0$

$y(t) = 0$

2°



2° $t > 0$

$$y(t) = \int_{0}^{t} h(\tau)\, x(t-\tau)\, d\tau$$

$$2^\circ \quad y(t) = \int_0^t e^{-\alpha\tau} \cdot 1 \, d\tau = \int_0^t e^{-\alpha\tau} \, d\tau = \frac{1}{\alpha} e^{-\alpha\tau} \Big|_0^t$$

$$= -\frac{1}{\alpha}\left(e^{-\alpha t} - e^{0}\right) = \frac{1}{\alpha}\left(1 - e^{-\alpha t}\right).$$

Detyra mik ka mate tiera!

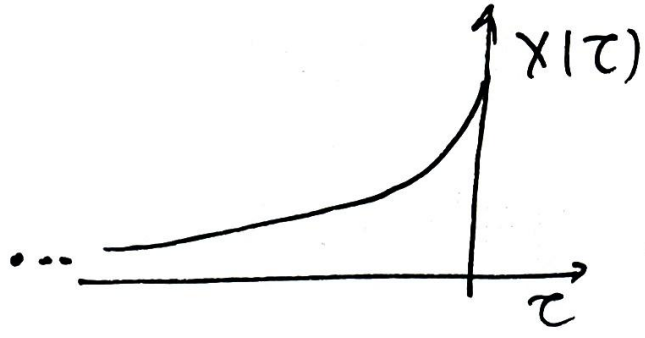
2.5 Gjeni daljen $y(t)$ për sinjalin kontinual nëse janë dhënë përgjigjja impulsive $h(t)$ dhe hyrja $x(t)$

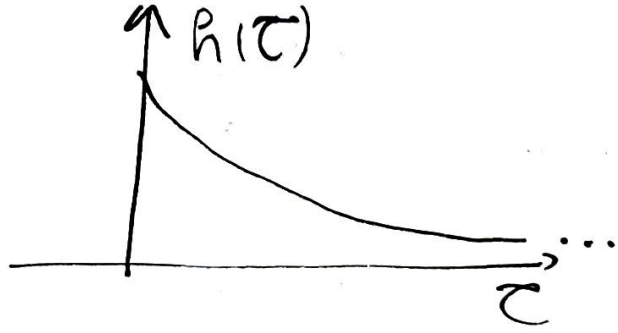
$$h(t) = e^{-\alpha t} u(t) \qquad x(t) = e^{\alpha t} u(-t)$$

$$\alpha = 2$$

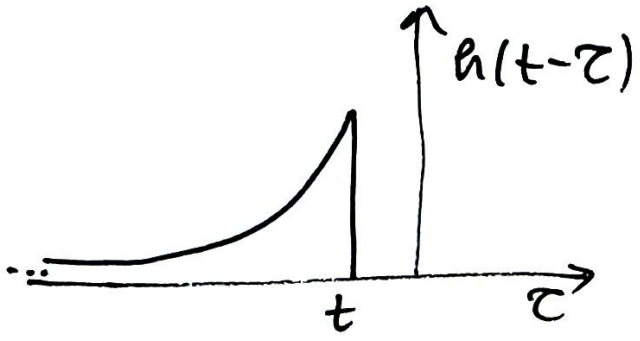
$$h(t) = e^{-2t} u(t) \qquad x(t) = e^{2t} u(-t)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) \, d\tau$$

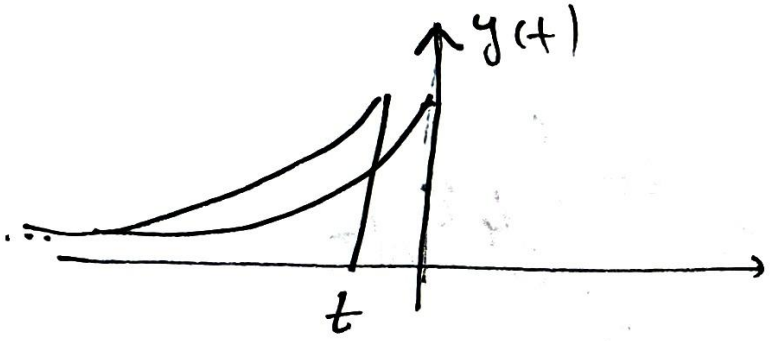




Pasqyrojmë dhe shtrendosim $h(t)$



1°



1° $t<0$

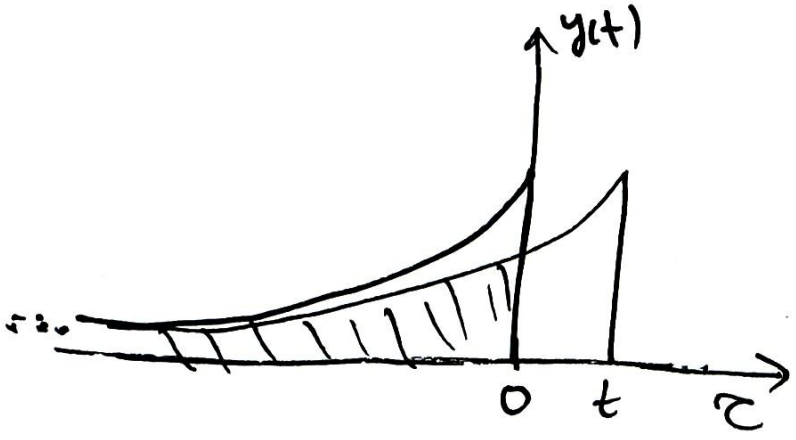
$$y(t)=\int_{-\infty}^{t} e^{\alpha\tau}\cdot e^{-\alpha(t-\tau)}\,d\tau$$

$$y(t)=\int_{-\infty}^{t} e^{\alpha\tau}\cdot e^{-\alpha t}\cdot e^{\alpha\tau}\,d\tau = e^{-\alpha t}\int_{-\infty}^{t} e^{2\alpha\tau}\,d\tau$$

$$= e^{-2t}\int_{-\infty}^{t} e^{4\tau}\,d\tau = e^{-2t}\,\frac{1}{4}\left[e^{4t}-e^{-\infty}\right]\to 0$$

$$= \boxed{\frac{1}{4}e^{2t}}$$

Rasti 2°



$t>0$

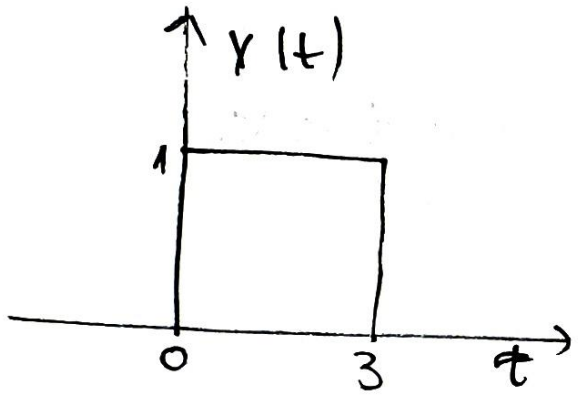
$$y(t) = \int_{-\infty}^{0} e^{-2t} e^{-2(t-\tau)} d\tau = e^{-2t} \int_{-\infty}^{0} e^{4\tau} d\tau$$

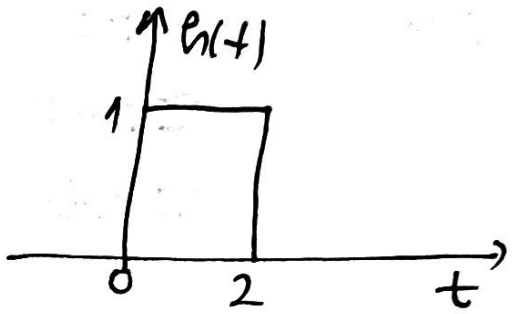
$$= e^{-2t} \frac{1}{4} \left(e^{4\cdot 0} - e^{-\infty} \right) = \frac{1}{4} e^{-2t}$$

26 Gjeni daljet $y(t)$ nëse janë dhënë hyrja dhe përgjigjja impulsive si në fig.

a) Metoda analitike

b) Metoda grafikone





a)

$$x(t) = u(t) - u(t-3) \qquad h(t) = u(t) - u(t-2)$$

$$x(\tau) = u(\tau) - u(\tau-3) \qquad h(t-\tau) = u(t-\tau) - u(t-\tau-2)$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} [u(\tau) - u(\tau-3)][u(t-\tau) - u(t-\tau-2)]\, d\tau$$

$$= \int_{-\infty}^{\infty} u(\tau)u(t-\tau)\, d\tau - \int_{-\infty}^{\infty} u(\tau)u(t-\tau-2)\, d\tau$$

$$- \int_{-\infty}^{\infty} u(\tau-3)u(t-\tau)\, d\tau + \int_{-\infty}^{\infty} u(\tau-3)u(t-\tau-2)\, d\tau =$$

1° $u(\tau)\,u(t-\tau)$

2° $u(\tau)\,u(t-\tau-2)$

3° $u(\tau-3)\,u(t-\tau)$

4° $u(\tau-3)\,u(t-\tau-2)$

1° $u(\tau)\,u(t-\tau)$

$\tau>0 \qquad t-\tau>0$

$-\tau>-t$

$\tau<t$

$0<t<2$

2° $u(\tau)\,u(t-\tau-2)$

$\tau>0 \qquad t-\tau-2>0$

$-\tau>2-t$

$\tau<t-2$

$0<\tau<t-2$

3° $u(\tau-3)\,u(t-\tau)$

$\tau-3>0 \qquad t-\tau>0$

$\tau>3 \qquad t<\tau$

$3<\tau<t$

4° $u(\tau-3)\,u(t-\tau-2)$

$\tau>3 \qquad \tau<t-2$

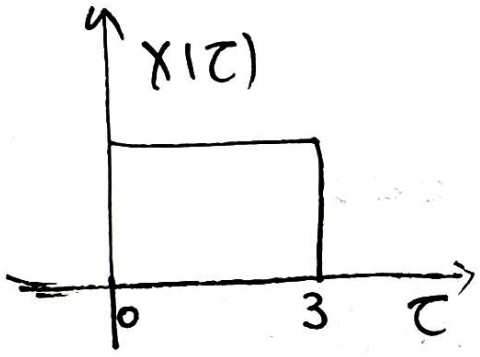
$3<\tau<t-2$

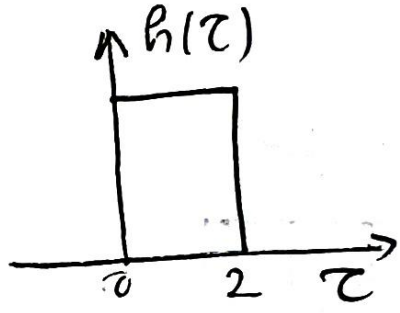
$$y(t)=\left(\int_0^t d\tau\right)u(t)-\left(\int_0^{t-2} d\tau\right)u(t-2)-\left(\int_3^t d\tau\right)u(t-3)$$

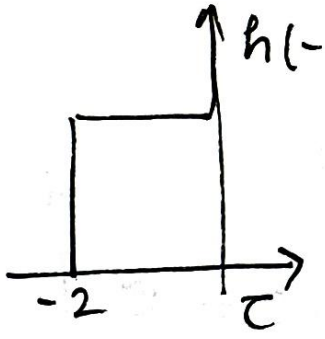
$$+\left(\int_3^{t-2} d\tau\right)u(t-5)$$

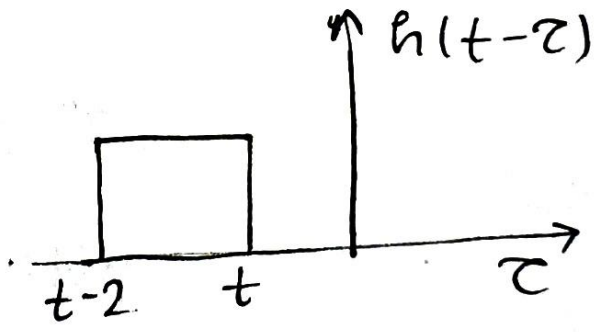
$$= t\,u(t)-(t-2)\,u(t-2)-(t-3)\,u(t-3)+(t-5)\,u(t-5)$$

b) Metoda grafikone





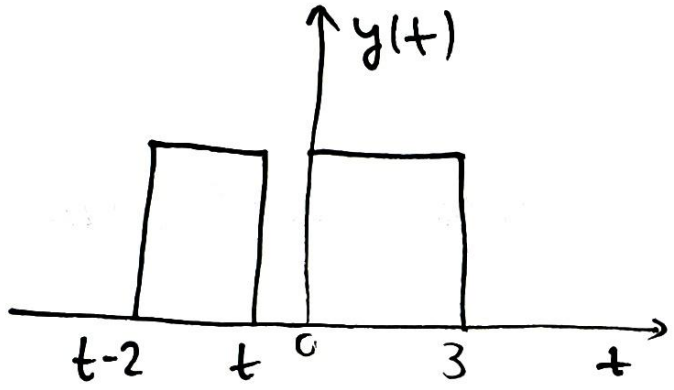




$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

Rastet:

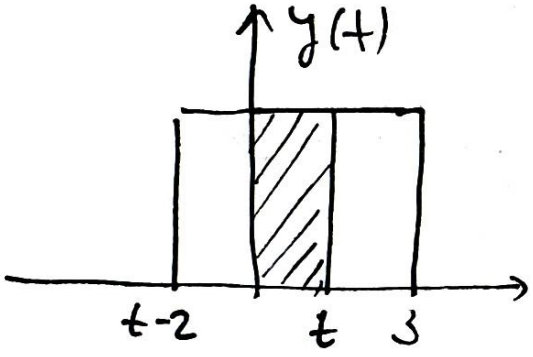
1°



1° $t<0$

$y(t)=0$

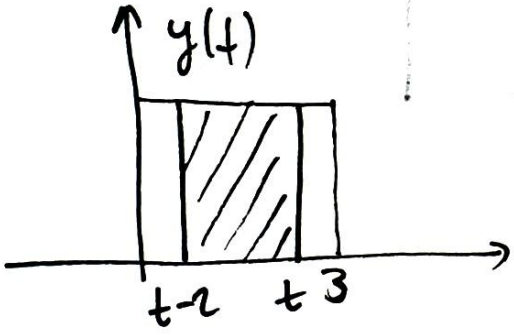
2°



2° $0<t<2$

$$y(t) = \int_0^t 1 \cdot 1\, d\tau = t$$

3°

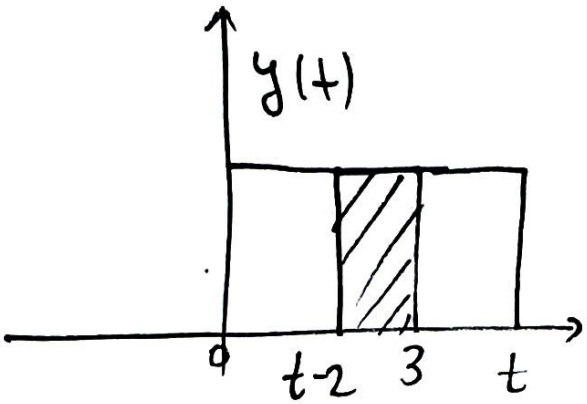


3° $2<t<3$

$$y(t) = \int_{t-2}^{t} 1 \cdot 1\, d\tau = \tau \Big|_{t-2}^{t}$$

$$= t - t + 2 = 2$$

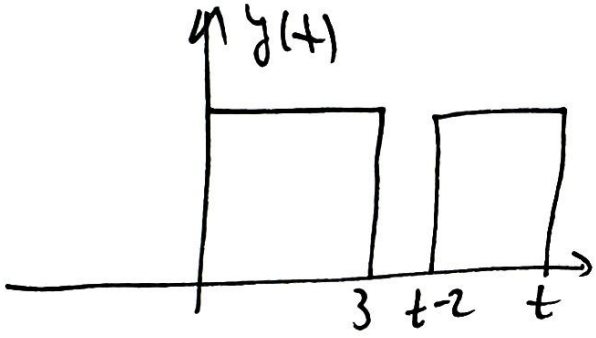
4°



$3 < t < 5$

$$y(t) = \int_{t-2}^{3} 1 \cdot 1 \, d\tau = 5 - t$$

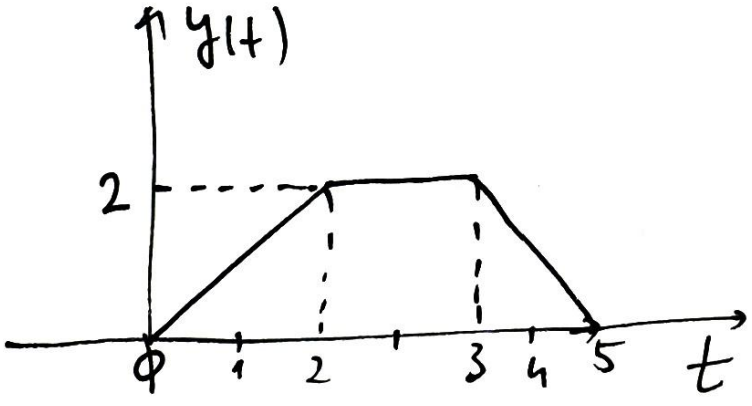
5°



$t > 5$

$y(t) = 0$

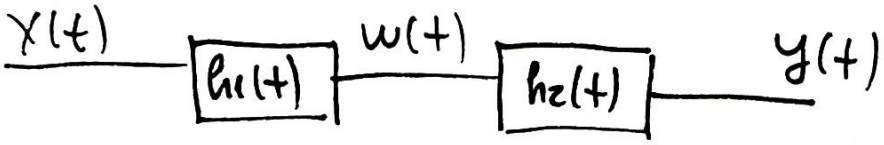
Figura $y(t)$

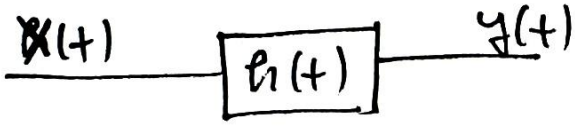


2.14 Sistemi në figurë është i formuar nga lidhja e dy sistemeve në seri, përcaktoni impulsivin të sistemit gjanë dhënë përmes $h_1(t)$ dhe $h_2(t)$

$$h_1(t) = e^{-2t}u(t) \qquad h_2(t) = 2e^{-t}u(t)$$

$$h(t) = ?$$





$$w(t) = x(t) * h_1(t)$$

$$y(t) = w(t) * h_2(t) = x(t) * h_1(t) * h_2(t)$$

$$= x(t) * h(t)$$

$$h(t) = h_1(t) * h_2(t)$$

$$h(t) = \int_{-\infty}^{\infty} h_1(\tau)\, h_2(t-\tau)\, d\tau = \int_{-\infty}^{\infty} e^{-2\tau} \cdot 2e^{-(t-\tau)}\, d\tau$$

$$= 2e^{-t}\int_{-\infty}^{\infty} e^{-\tau}\, d\tau = 2e^{-t}\int_{0}^{t} e^{-\tau}\, d\tau$$

kufijt

$\tau > 0$ $\quad t-\tau > 0$

$-\tau > -t$

$\tau < t$

$$= 2e^{-t}\left[-e^{-\tau}\right]_0^t = 2e^{-t}\left[-e^{-t}+1\right] = 2e^{-t}\left[1-e^{-t}\right]$$